# EXPLORATORY DATA ANALYSIS

# TITANIC PASSENGER DATASET

A Python-based study of passenger characteristics and survival patterns

| | |
|---|---|
| **Dataset** | Titanic training dataset (train.csv) |
| **Records** | 891 passenger records |
| **Attributes** | 12 columns |
| **Tools** | Python • Pandas • NumPy • Matplotlib • Seaborn |
| **Visualisations** | 8 charts from the executed Google Colab notebook |

Prepared from the supplied notebook and dataset

# 1. Introduction

Exploratory Data Analysis (EDA) is the process of examining a dataset using summary statistics and visualisations to understand its structure, identify missing values, and discover patterns. This report explores the Titanic passenger dataset, focusing on survival outcomes and their relationship with passenger gender, passenger class, age, and fare.

# 2. Objectives

• Understand the structure and size of the dataset.
• Identify missing values and duplicate records.
• Summarise the distribution of survival outcomes.
• Explore survival patterns by gender and passenger class.
• Examine age and fare distributions and relationships.
• Use correlation and pairwise plots to explore numerical variables.

# 3. Dataset Description

The supplied dataset contains **891 rows** and **12 columns**. The columns describe passenger identifiers, survival status, ticket class, name, sex, age, family relationships aboard, ticket, fare, cabin, and port of embarkation. The target-like field **Survived** is coded as 0 for did not survive and 1 for survived.

# 4. Data Quality Check

| Column | Missing values |
|---|---|
| Age | 177 |
| Cabin | 687 |
| Embarked | 2 |

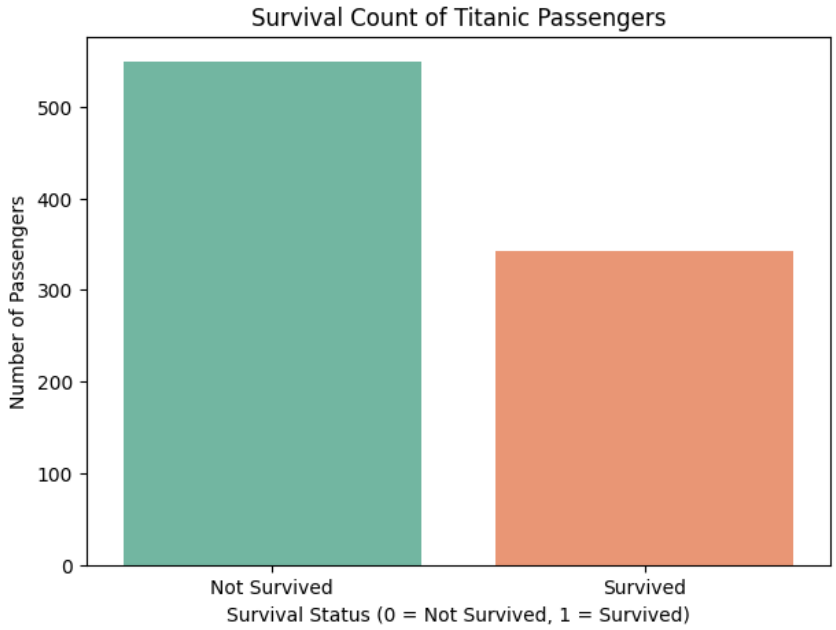
There are **0 duplicate rows**. Age has 177 missing values, Cabin has 687, and Embarked has 2. These missing values should be considered before any further analysis or modelling. No imputation or row removal is assumed in this report.

# 5. Summary Statistics

• Average passenger age (excluding missing values): 29.70 years.
• Average ticket fare: 32.20.
• Passengers who did not survive: 549.
• Passengers who survived: 342.
• Overall survival rate: 38.38%.

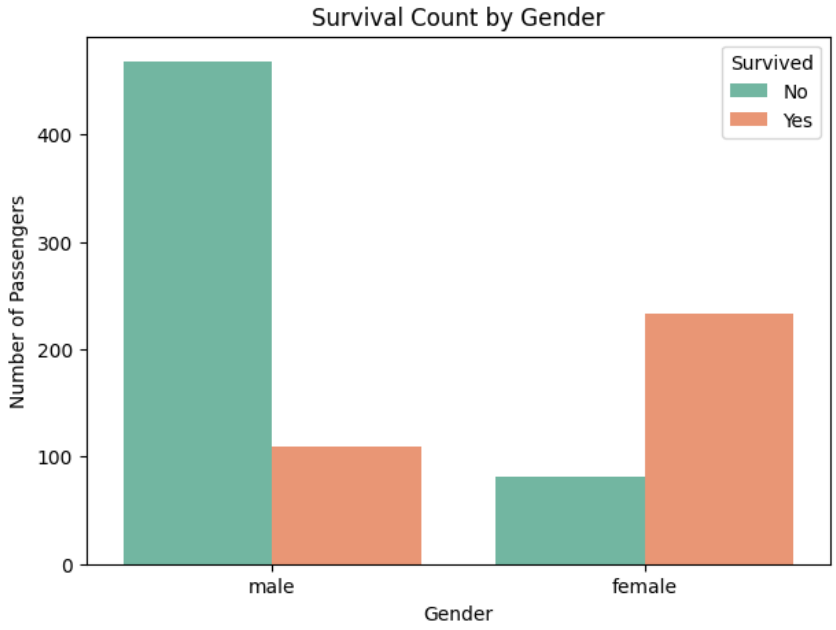
# 6. Visual Analysis and Observations

## 6.1 Survival Count



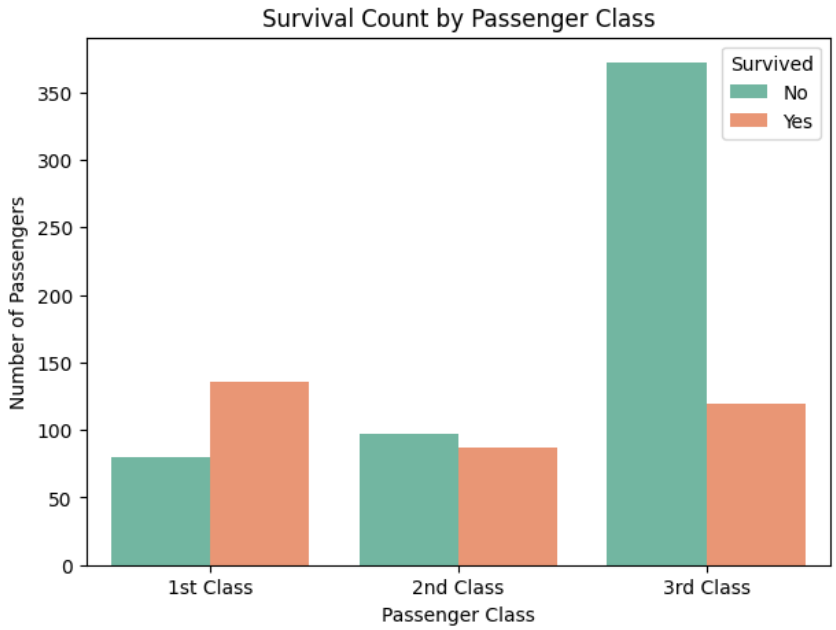

**Observation:** The dataset contains 549 passengers who did not survive and 342 who survived. The number of non-survivors is higher; the overall survival rate is 38.38%.

## 6.2 Survival Count by Gender



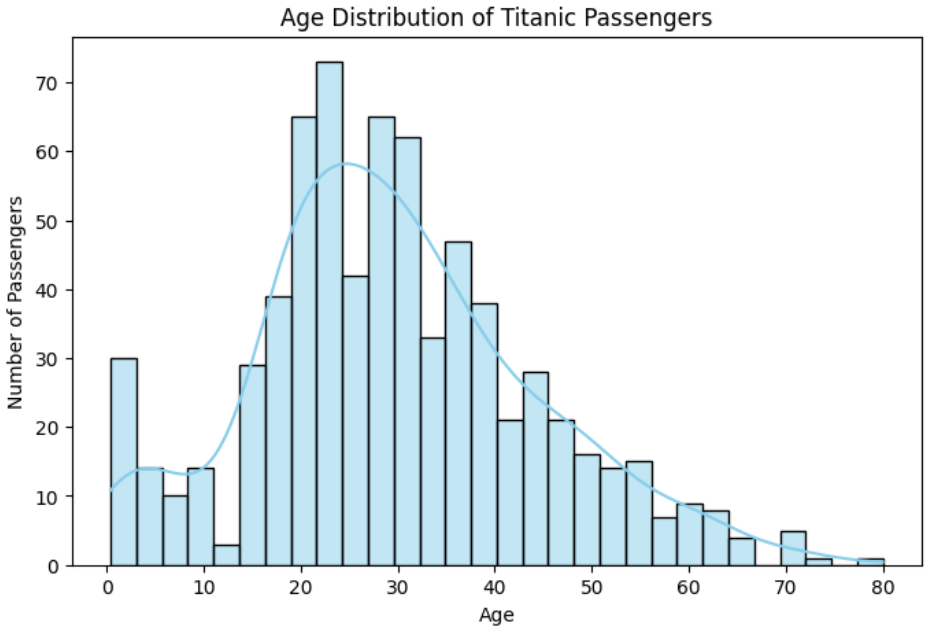

**Observation:** Female survival rate is 74.20%, while male survival rate is 18.89%. The chart shows a clear difference in survival outcomes by gender in this dataset.

## 6.3 Survival Count by Passenger Class



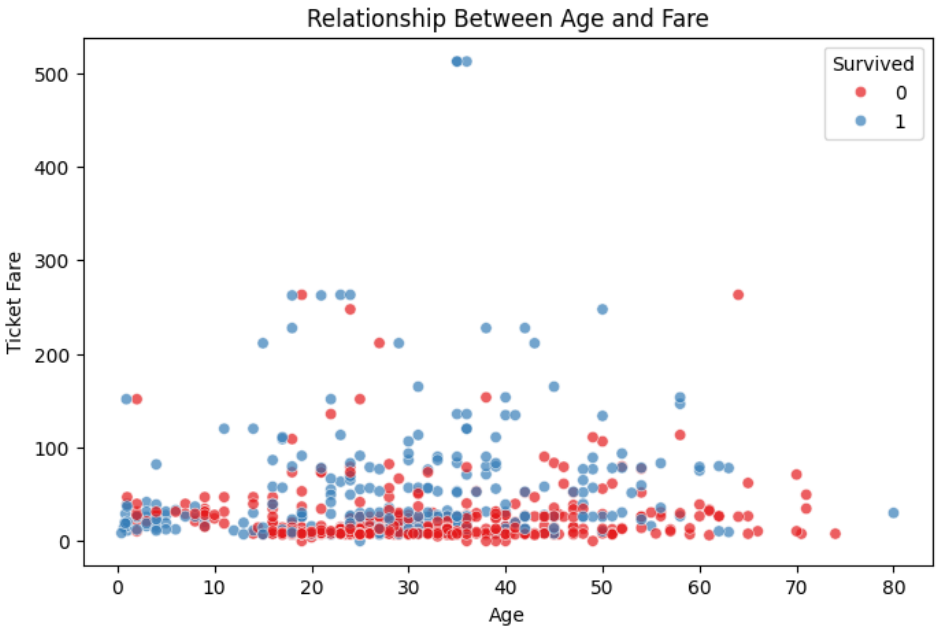

**Observation:** Survival rates are 62.96% for 1st class, 47.28% for 2nd class, and 24.24% for 3rd class. The first-class group has the highest survival proportion and the third-class group the lowest.

## 6.4 Age Distribution of Titanic Passengers



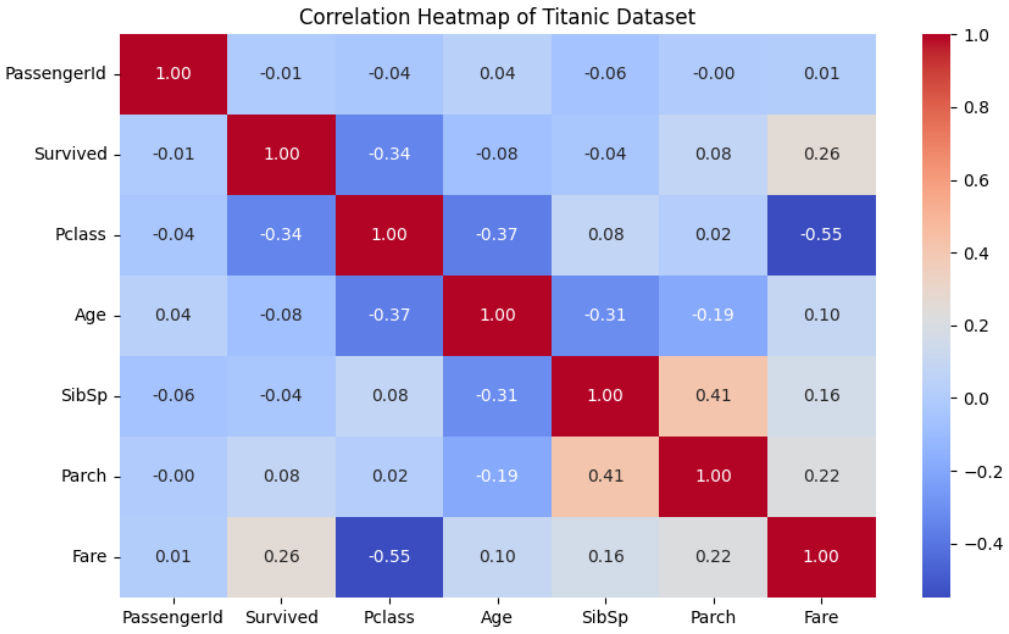

**Observation:** The average recorded age is 29.70 years. The histogram displays the distribution of recorded ages; missing age values are not shown as age observations in the histogram.

## 6.5 Age vs Fare



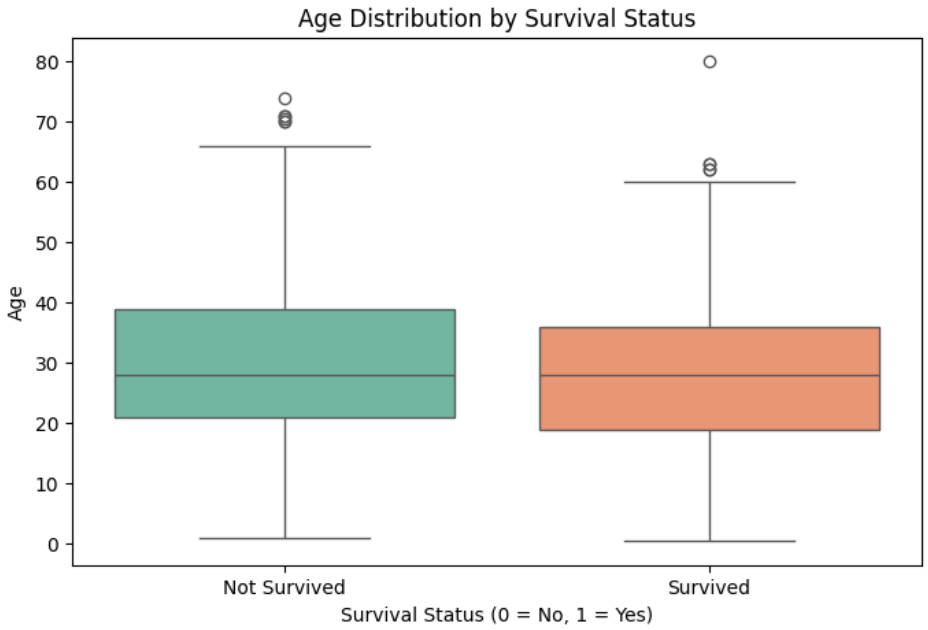

**Observation:** The scatterplot compares passenger age and fare, with colour indicating survival. Fares are concentrated at lower values, while a smaller number of passengers paid much higher fares. The points show that age and fare alone do not neatly separate all survival outcomes.

## 6.6 Correlation Heatmap



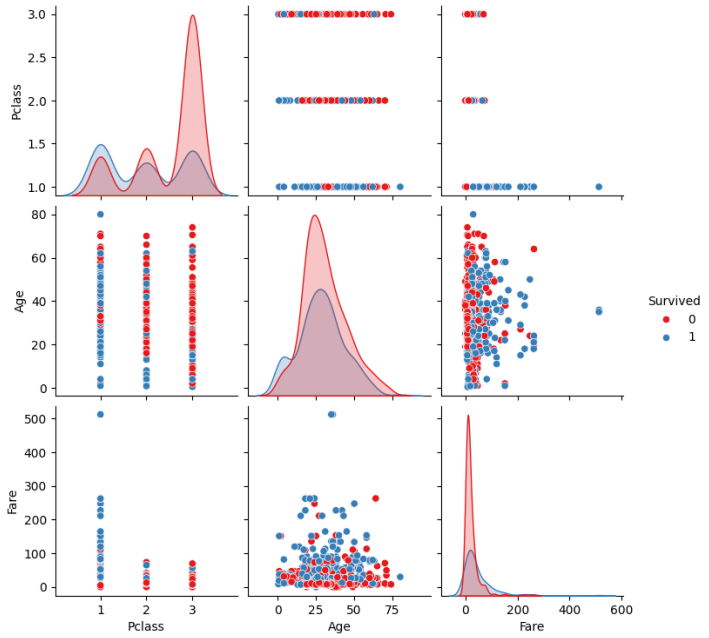

**Observation:** The numerical correlation with Survived is approximately -0.338 for Pclass, 0.257 for Fare, and -0.077 for Age. These are associations, not evidence of causation.

## 6.7 Age by Survival (Boxplot)




**Observation:** The median recorded age is 28 years for both survival groups. The age distributions overlap, suggesting that age alone does not clearly distinguish survivors from non-survivors.

## 6.8 Pairplot of Selected Numerical Variables




**Observation:** The pairplot presents pairwise relationships among Survived, Pclass, Age, and Fare, with points grouped by survival status. It provides a compact overview of patterns and overlap across these variables.

# 7. Key Findings

• More passengers did not survive (549) than survived (342).
• The overall survival rate in the dataset is 38.38%.
• Female passengers had a higher survival rate than male passengers in this dataset.
• First-class passengers had the highest survival rate among the three passenger classes; third-class passengers had the lowest.
• The average recorded age was 29.70 years, with 177 age values missing.
• Fare has a positive correlation with survival (0.257), while Pclass has a negative correlation (-0.338). Correlation does not establish causation.

# 8. Conclusion

The EDA of the Titanic dataset revealed differences in survival outcomes across gender and passenger class. Female passengers and passengers travelling in first class had higher survival proportions in the supplied data. The age and fare visualisations provided additional context, while the correlation heatmap and pairplot helped examine relationships among numerical variables. The analysis also identified missing values, particularly in the Cabin and Age columns. Overall, visualisation and summary statistics provide a useful first understanding of the dataset before any further data cleaning or predictive analysis.

# Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, and Google Colab.

Report prepared using the supplied EDA notebook and Titanic CSV dataset. All eight charts are taken from the notebook outputs.